KERAAL Research Direction & This Week's Plan

Subject-Independent Rehabilitation Exercise Error Classification · Prepared 2026-09-26

TL;DR

This week's goal is to **secure the data, confirm the label structure, and freeze the evaluation splits**. Model training waits until next week. Five key findings from Paper 2 and the dataset:

1. **We need to resolve a scope conflict first.** The proposal puts body-region localization out of scope, while the project-idea document makes "Error type + Body region" the core.
2. **Body-part labels exist only in Groups 1A/2A (about 300 recordings), not in Group 3, our training set.** There are only four values (LeftArm / RightArm / BothArms / Spine), so they may be almost fully determined by the error type.
3. **Paper 2's accuracy numbers are hard to take at face value because of class imbalance.** The torso-rotation test set has 72 Correct out of 82, so predicting Correct for everything gives 87.8%, higher than the reported 73.17%. This is exactly why our proposal uses Macro-F1.
4. **Group 3 has only three healthy participants.** A subject-disjoint split inside Group 3 is at most 3-fold, so "train on Group 3 → test on 1A+2A" is the natural main protocol.
5. **Download order: 2A (338 MB) → 1A (1.4 GB) → 3 (8.2 GB).** 1B (13 GB) and 2B have no labels and are not needed now.

Paper 2: Marusic et al. (ICORR 2025)

Paper 2 shows that error-type classification on KERAAL is feasible, but its evaluation has clear gaps, and those gaps are where we can contribute. (paper, code)

What they did

- **Input:** 3D positions of the 25 Kinect V2 joints only (quaternion orientations unused). Each recording is interpolated to T frames.
- **Model:** an adapted HyperFormer (hypergraph + 10 self-attention layers). Joints are split into six subgraphs, with extra emphasis on the arms.
- **Training:** one model per exercise, 4 classes (Correct + Error1–3), Adam, lr 2.5×10^{-3}, batch 10, 600 epochs.
- **Explanation:** joint importance from averaged self-attention maps, comparing correct vs. incorrect executions.

Evaluation scenarios

Scenario	Train	Test	Subject-disjoint?
S1	Group 3 (healthy, simulated errors)	Groups 1A + 2A	Yes (different people)
S2	80% of all data (random, label-stratified)	Remaining 20%	No (random by recording)
S3	85% of healthy-participant data	Patients (G1) + 15% of healthy	Partly (healthy data mixed)

S1 results (reported accuracy)

Exercise	Transformer	LSTM best	LSTM mean	GMM
Torso rotation	73.17%	64.44%	53.89%	27.78%

Exercise	Transformer	LSTM best	LSTM mean	GMM
Flank stretch	64.10%	43.04%	31.64%	25.32%
Hiding face	74.28%	56.19%	49.1%	33.33%

S2 and S3 are shown only as confusion-matrix figures, with no numbers.

Insights for us

1. **The model may not beat the majority baseline.** The paper itself states the torso-rotation test set is Correct 72 / Error1 4 / Error2 6 / Error3 0. Predicting Correct everywhere gives 72/82 = 87.8%. → We always report the **majority baseline, Macro-F1, and balanced accuracy** together.
2. **The "better performance" in S2 may be subject leakage.** Group 3 has only three people, so a random 80:20 split puts repetitions of the same person in both train and test. → Quantifying the gap between random and subject-disjoint splits is a contribution in itself.
3. **The real challenge is sim-to-real transfer.** Group 3 consists of healthy people deliberately acting out errors; the test set contains patients' natural errors. Whether normalized kinematic features narrow this gap maps directly onto our hypothesis.
4. **Flank stretch "Body is not tilted" is the hardest error.** The original paper blames Kinect precision. → Error-specific features such as lateral trunk tilt (shoulder-line and pelvis-line angles, spine-vector tilt) are a concrete target to test.
5. **Joint importance: arms matter, legs matter less.** → An "upper body only" ablation is cheap. The paper never checked whether attention importance agrees with the annotated bodyPart.
6. **Unreported processing rules:** how Incomplete / Motionless / "Other" were handled, and whether Small vs. Large errors were distinguished. Stating these makes our work reproducible.
7. **For hiding face, Error1 and Error3 have identical descriptions in the paper ("Arms are not raised enough").** We need to check the real meaning in the original labeling guide (a French PDF).

KERAAL Dataset Review

KERAAL is essentially the only public dataset that fits our task, but the labeled patient data is small (about 300 recordings) and imbalanced. The facts below come from the official documentation, papers and annotation spec; items to verify against the actual files are listed at the end.

Groups and how we use them

Group	Who	Particip.	Recordings	Labels	Size	Our use
3	Healthy, simulated errors	3	540	Error labels only (no bodyPart)	8.2 GB	**Train**
1A	Patients, robot-coached	6	249	Full (evaluation, error, severity, bodyPart, time span)	1.4 GB	**Test**
2A	Healthy, robot-coached	6	51	Full	338 MB	**Test**
1B	Patients	12	1,631	None	13 GB	Not needed (only for self-supervised pretraining)
2B	Healthy	6	151	None	1.3 GB	Not needed

Sources: KERAAL official page, KERAAL paper. License: data CC BY-NC-SA, code BSD-2.

Exercises and error definitions

Exercise (code)	Error1	Error2	Error3
Torso rotation (RTK)	Arms not raised enough	Torso rotation not sufficient	Body leaned to the side
Flank stretch (ELK)	Opposite arm not along the body	Body not tilted	Upper arm not bent
Hiding face (CTK)	Arms not raised enough	Arms not outspread enough	Same wording as Error1 in the paper (verify)

Annotation scheme (keraalSpec.xml, Anvil XML)

Track / attribute	Values
Global evaluation	Correct, Incorrect, Incomplete, Motionless
Global error → type	none, Error1, Error2, Error3, Other
Global error → Evaluation (severity)	SmallError, LargeError
Global error → bodyPart	LeftArm, RightArm, BothArms, Spine
Temporal error	Same attributes + start/end time

Inter-annotator agreement is Cohen's κ = 0.63, so label noise should be listed as a limitation. (spec file)

Skeleton formats

Source	Joints	Values/frame	Groups	Notes
Kinect V2	25	175 (3 position + 4 quaternion)	All	**Our default input**, ASCII table
Vicon	17 markers	119	Group 3 only	Unsuitable: absent from test set
BlazePose	33	3D	All	Video-based, for side experiments
OpenPose	14	2D	All	Excluded (2D)

Source: readme_files_format.txt

Risks and what to verify in the files this week

- ☐ **Participant ID rule:** can subject IDs be parsed from file names? Without them, subject-disjoint splits are impossible.
- ☐ **Class counts:** per-class counts in 1A+2A. According to the paper, some classes have only 0–6 examples.
- ☐ **Group 3 composition:** is 540 = 3 people × 3 exercises × 4 classes × 15 reps, i.e. balanced? (estimate, unverified)
- ☐ **bodyPart × error cross-tab:** is bodyPart essentially determined by error type? If so, multi-task learning adds little new information.
- ☐ **Kinect frame rate and column order:** fps is not documented; check via timestamps before computing velocity.
- ☐ **Missing matches:** recordings with annotations but no skeleton (or vice versa), and dropped frames (all-zero rows).

Our Research Direction

We recommend making this the core question and treating body region as exploratory analysis: **Under subject-independent, simulated-to-real evaluation, do kinematic features aligned with the error definitions help?** This keeps the proposal's scope while cheaply preserving the project-idea goal of telling users where the error is.

Why body-region multi-task is not the core

- Group 3 (training) has no bodyPart labels, so multi-task learning could only use the error samples in 1A/2A (a few dozen).
- bodyPart has only four values and is likely implied by the error type (e.g. "arms not raised" → BothArms, "insufficient rotation" → Spine).
- The informative part is mainly **LeftArm vs. RightArm** (side), which left-right symmetry features could also resolve with simple rules.
- If this week's bodyPart × error cross-tab shows enough independent information, we can promote multi-task back to the core.

Research questions

RQ	Question	Comparison
RQ1 (core)	Do normalized kinematic features improve subject-independent Macro-F1 over raw coordinates?	raw vs. angles vs. angles+velocity vs. +symmetry (feature-group ablation)
RQ2	Does a TCN match or beat an LSTM?	LSTM vs. TCN, same input and splits
RQ3	How much does random splitting inflate performance?	recording-level random vs. subject-disjoint
RQ4 (exploratory)	Do the features/joints the model relies on agree with annotated bodyPart?	feature-group importance vs. 1A/2A bodyPart

Evaluation protocol

1. **P1 (main):** train on Group 3 → test on 1A + 2A. Same setting as Paper 2's S1, so directly comparable; we also report accuracy for reference.
2. **P2:** leave-one-subject-out within Group 3 (3 folds), for subject generalization within one distribution.
3. **P3 (for RQ3):** split the same data 80:20 randomly and 80:20 grouped by subject, and measure the gap.
4. **Common to all protocols:** 4 classes per exercise, majority baseline, a simple statistical-feature + Random Forest baseline, mean ± SD over 5 seeds, Macro-F1 (primary) + balanced accuracy + per-class recall + confusion matrices.
5. **Explicit label rules:** by default, exclude Incomplete / Motionless / Other and count both Small and Large errors as errors (with a separate per-severity table).

Draft error-specific features

Error	Candidate features
Arms not raised enough (RTK-E1, CTK-E1)	Shoulder elevation angle (trunk vector vs. upper-arm vector), wrist height / body height
Insufficient torso rotation (RTK-E2)	Horizontal-plane angle between shoulder line and pelvis line
Body leaned / not tilted (RTK-E3, ELK-E2)	Frontal-plane tilt of the spine vector (SpineBase→Neck)
Arm not bent (ELK-E3)	Elbow flexion angle

Error	Candidate features
Arms not outspread enough (CTK-E2)	Wrist-to-wrist distance / shoulder width, shoulder abduction angle

Roshini will finalize this mapping after checking the error-definition document this week; it then links directly to the demo's feedback cues (e.g. "Raise your arms a little higher").

Step-by-Step Plan This Week (9/28–10/2)

By Friday we want **master_index.csv, class-count tables, frozen split files, and a 1–2 page data report**. No models this week. The inventory script is shared as keraal_inventory.py.

Roles

Member	Role this week
Hyungeon (AI & DS)	Data pipeline lead: run the inventory, master_index.csv, splits.json
Uijin (Mechanical Control & DS)	Skeleton loader, coordinate system and fps checks, motion GIFs, normalization prototype
Roshini (AI & Biology)	Finalize error definitions (French labeling guide), error → feature → feedback mapping, interpret bodyPart
Joshi (first-year)	Downloads and storage, sample exploration, prior-work results table, meeting notes

Daily plan

Day	Goal	Tasks (owner)	Done when
Mon 9/28	Kickoff + start downloads	30-min kickoff to agree on a draft scope (all). Get the sample zip, readme and group2A; download 1A and 3 overnight (Joshi). Start translating the labeling PDF (Roshini). Set up repo and environment (Hyungeon). Document Kinect V2 joint order and coordinate system (Uijin)	2A is in shared storage
Tue 9/29	Understand file structure	List tar contents to learn folder and file-name conventions; fix the participant-ID regex (Hyungeon + Joshi). First inventory run on 2A (Hyungeon). Open one file, check 175 columns and fps (Uijin)	Draft master_index for 2A; participant IDs parsed for 100% of files
Wed 9/30	Organize all data + eyeball it	Extract 1A and 3, run the full inventory (Hyungeon). 12 GIFs from Group 3: 3 exercises × 4 classes (Uijin). Check that each GIF matches its error definition (Roshini)	12 GIFs + review notes
Thu 10/1	Tables + freeze splits	Class and per-participant counts, bodyPart × error cross-tab, majority baseline, data-quality issue list (Hyungeon + Roshini). Save P1/P2/P3 splits with fixed seeds as splits.json (Hyungeon). Prior-work results table (Joshi)	splits.json committed; 4 tables
Fri 10/2	Team review & decisions	Present the data report, settle the decisions below, plan next week (preprocessing + RF baseline + LSTM)	Decisions recorded; next week's owners assigned

Download tips

- Most of the size is video. First run `tar -tJf group3.tar.xz | head -100` to see folder names, then extract only Kinect and annotation files to save disk space (adjust `--wildcards` patterns to the real folder names).
- Keep the original tar files and record their sizes so the paper can state the exact data version.
- The data is CC BY-NC-SA: non-commercial research use only, and never push the raw data to GitHub.

End-of-week checklist

- ☐ Kinect + annotation files for 1A, 2A and 3 are in shared storage
- ☐ master_index.csv (per recording: group, subject, exercise, evaluation, error, severity, bodyPart, kinect_path, n_frames)
- ☐ Class-count table, per-participant table, bodyPart × error cross-tab, majority baseline
- ☐ 12 GIFs (exercise × class)
- ☐ Error definition → feature → feedback-cue mapping table
- ☐ splits.json (P1, P2, P3; fixed seeds)
- ☐ 1–2 page data report (becomes the draft of the paper's Dataset section)

Decisions for the Team Meeting

#	Decision	Proposal	Rationale
1	Is body region part of the core?	Exploratory (RQ4); revisit after Thursday's cross-tab	No bodyPart in Group 3; likely tied to error type
2	Main evaluation protocol	P1: Group 3 → 1A+2A	Directly comparable to Paper 2 S1; subject-independent
3	Label handling	Exclude Incomplete/Motionless/Other; count Small+Large as errors	Makes explicit what Paper 2 left unreported
4	Skeleton source	Kinect V2 (25 joints)	Available for all groups; same as Paper 2
5	Model granularity	One 4-class model per exercise	Comparable to Paper 2; error definitions differ per exercise
6	Data storage	Lab server or shared drive (about 10 GB)	Everyone works from the same files

Sources

- Marusic et al., Skeleton-Based Transformer for Classification of Errors (arXiv 2504.13866)
- hyperphysio code
- Nguyen et al., KERAAL dataset paper (arXiv 2407.00521)
- KERAAL official page (download links)
- readme_files_format.txt (file formats)
- KeraalDataset GitHub
- keraalSpec.xml (annotation spec)
- Nguyen, Low-Back Pain Physical Rehabilitation by Movement Analysis in Clinical Trial (arXiv 2601.06138)